

Climate 5000 Series - Gen 4

18K BTU Single Zone Regular System

4-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

18K Btu Single Zone Regular System - 4-Way Cassette IDU

Indoor part number 8-733-962-718

Indoor model number BMS500-AAU018-1AHCXD

Outdoor part number 8-733962-686

Outdoor model number BMS500-AAS018-1CSXRD

Efficiency Data AHRI 210/240 2023

SEER2 20.7

EER2 12.5

HSPF2 Region IV 11.4

HSPF2 Region V 8.5

COP(47F) 3.36

COP (5F) 2.20

Performance Data AHRI 210/240 2023

Capacity Range 5300~17800

Rated Capacity 16800

Current 6.1

Input Power 1344

Capacity Range 7100~21500

Rated Capacity (47F) 18000

Rated Capacity (5F) 12800

Current 7.2

Input Power 1570

Electrical Data

Power supply (V / Ø / Hz) 208-230 / 1 / 60

Min. circuit ampacity indoor/outdoor (A) 3.0 / 18.0

Max. fuse indoor/outdoor (A) 15.0 / 20.0

Compressor Data

Type: Rotary

Refrigerant oil /oil charge (ml) VG74 / 440

Input (W) 1090

Rated current (RLA/A) 7.2

Piping

Refrigerant precharge (oz/kg) 47.97 / 1.36

Length for precharge (ft/m) 24.6 / 7.5

Liquid side (in/mm) 1/4" / Ø6.35

Gas side (in/mm) 1/2" / Ø12.7

Additional Charge of Refrigerant (oz/ft)/(g/m) 0.16 / 15

Max. refrigerant pipe length (ft/m) 98.40 / 30

Max. difference in height (ft/m) 65.60 / 20

Operation Range

ODU Cooling (°F / °C) -13~122 / -25~50

Heating (°F / °C) -13~75 / -25~24

IDU Cooling (°F / °C) 60~90 / 16~32

Heating (°F / °C) 32~86 / 0~30

System Data

Refrigerant type R454B

Design pressure (PSIG) 550 / 340

Dehumidification (pts/hr) 4.33

Fan motor model ZKFN-45-8-2

Fan motor qty (ea) 1

Fan motor input (W) 50.0

Fan speed (RPM) 800 / 750 / 700

Air flow, Turbo / H / M / L / Si (CFM) 447.3 / 388.5 / 317.8 / 194.2 / 147.2

Approx. Air Throw Range ft (m) 9.8 / 3

Noise level, Turbo / H / M / L / Si (dB(A)) 48 / 44.0 / 41.0 / 28.5 / 24.5

Net weight (lbs / kg) 35.27 / 16.0

Gross weight (lbs / kg) 41.89 / 19.0

Panel net weight (lbs / kg) 5.95 / 2.7

Gross panel weight (lbs / kg) 9.48 / 4.3

Drain pipe diameter (mm) ODØ25mm

Fan motor model ZKFN-80-10-1L

Fan motor qty (ea) 1

Fan motor input (W) 161.0

Fan motor speed (RPM) 780 / 700 / 550

Air flow, Max (CFM) 1765.8

Noise level, Max (dB(A)) 57.5

Net weight (lbs / kg) 92.59 / 42.0

Gross weight (lbs / kg) 100.31 / 45.5

Accessories

BMS-CCG-XXD (8733962708) Decorative Grille - 9K ~ 18K

BMS-WT2-XXD (8733962707) Wired Wall Thermostat

BMS-DRY-XXD (8733969263) 24V Interface

BMS-PK2-7X1 (8733951011) 25ft Lineset

Standard Parts

8-733-969-065 Remote Control

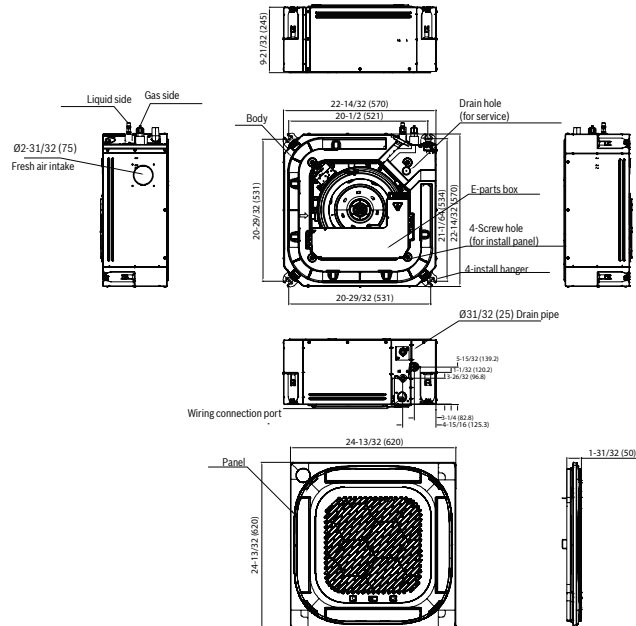
Climate 5000 Series - Gen 4

18K Btu Single Zone Regular System - 4-Way Cassette IDU



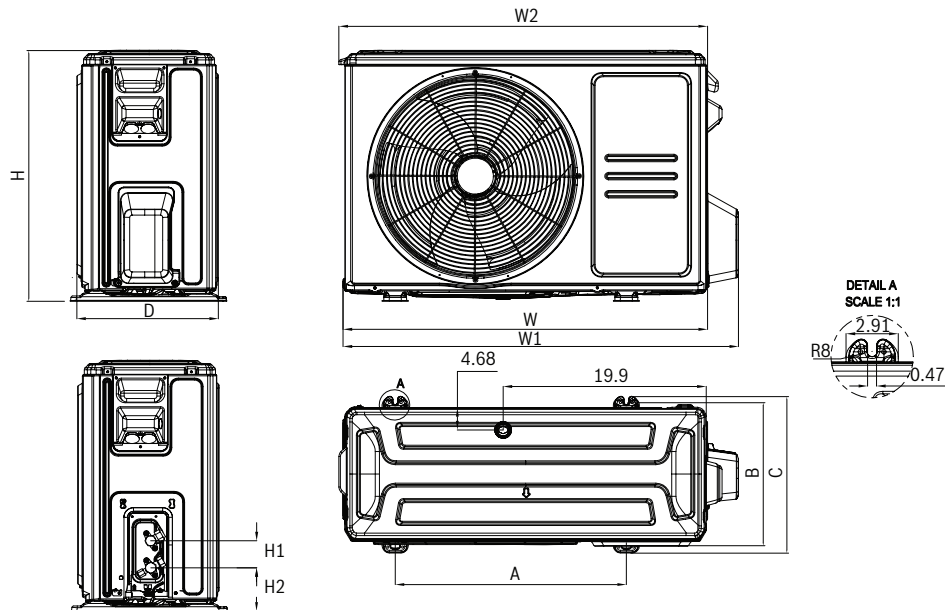
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Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU018-1AHCXD	22.44	22.44	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS018-1CSXRD	35	13.46	26.5	37.6	35.24	26.1	13.7	14.96	2.36	4.25

Climate 5000 Series - Gen 4

18K Btu Single Zone Regular System - 4-Way Cassette IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 447 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
-13°F (-25.0°C)	TC	13466	13484	13498	13499	13517	14277	14294	14308	14311	14328	15130	15150	15167	15169	15186	16042	16059	16077	16079	16099
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.59	0.59	0.59	0.59	0.59	0.63	0.63	0.63	0.63	0.63	0.66	0.67	0.67	0.67	0.67	0.70	0.70	0.71	0.71	0.71
-4°F (-20.0°C)	TC	14501	14521	14537	14537	14557	15375	15394	15409	15412	15430	16294	16315	16333	16336	16355	17276	17295	17313	17316	17338
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	0.61	0.61	0.61	0.61	0.61	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73
0°F (-17.8°C)	TC	14962	14982	14998	14999	15019	15864	15882	15898	15901	15920	16811	16833	16852	16855	16874	17824	17844	17863	17866	17888
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.62	0.62	0.62	0.62	0.62	0.66	0.66	0.66	0.66	0.66	0.70	0.70	0.70	0.70	0.70	0.74	0.74	0.74	0.74	0.74
5°F (-15.0°C)	TC	15537	15558	15575	15576	15597	16474	16493	16509	16513	16532	17458	17481	17500	17503	17523	18510	18530	18550	18553	18576
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	0.63	0.63	0.63	0.63	0.64	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	0.75	0.76	0.76	0.76	0.76
14°F (-10.0°C)	TC	15735	15756	15773	15774	15795	16683	16703	16720	16723	16743	17680	17703	17723	17726	17746	18745	18766	18786	18789	18812
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.62	0.62	0.62	0.62	0.62	0.66	0.66	0.66	0.66	0.66	0.70	0.70	0.70	0.70	0.70	0.74	0.74	0.74	0.74	0.74
17°F (-8.3°C)	TC	15801	15822	15839	15840	15861	16753	16773	16790	16793	16813	17754	17777	17797	17800	17820	18824	18844	18865	18868	18891
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.62	0.62	0.62	0.62	0.62	0.65	0.65	0.66	0.66	0.66	0.69	0.69	0.69	0.69	0.70	0.73	0.74	0.74	0.74	0.74
23°F (-5.0°C)	TC	15944	15966	15983	15984	16005	16905	16925	16942	16945	16965	17915	17938	17958	17962	17982	18995	19015	19036	19039	19063
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.62	0.62	0.62	0.62	0.62	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.74
35°F (1.7°C)	TC	16231	16253	16270	16271	16293	17209	17230	17247	17250	17271	18237	18261	18281	18285	18305	19336	19357	19378	19381	19406
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.61	0.61	0.61	0.61	0.62	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73
41°F (5.0°C)	TC	16039	16060	16078	16079	16100	17006	17026	17042	17046	17066	18021	18045	18065	18068	18089	19107	19128	19149	19152	19176
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	0.65	0.65	0.65	0.65	0.65	0.68	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73	0.77	0.77	0.77	0.77	0.77
47°F (8.3°C)	TC	15847	15868	15885	15886	15907	16802	16822	16838	16842	16862	17806	17829	17849	17852	17872	18878	18899	18919	18922	18946
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.68	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72	0.76	0.76	0.76	0.76	0.76	0.81	0.81	0.81	0.81	0.81
50°F (10.0°C)	TC	15852	15873	15890	15891	15912	16807	16827	16843	16847	16867	17811	17834	17854	17857	17877	18884	18905	18925	18928	18952
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.70	0.70	0.70	0.70	0.70	0.74	0.74	0.74	0.74	0.74	0.78	0.78	0.79	0.79	0.79	0.83	0.83	0.83	0.83	0.83
59°F (15.0°C)	TC	15867	15888	15905	15906	15927	16823	16843	16859	16863	16883	17828	17851	17871	17874	17894	18902	18922	18943	18946	18970
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.75	0.76	0.76	0.76	0.76	0.80	0.80	0.80	0.80	0.80	0.85	0.85	0.85	0.85	0.85	0.90	0.90	0.90	0.90	0.90
68°F (20.0°C)	TC	15881	15903	15920	15921	15942	16838	16858	16875	16878	16898	17844	17868	17888	17891	17911	18920	18940	18961	18964	18987
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	0.81	0.81	0.81	0.81	0.81	0.86	0.86	0.86	0.86	0.86	0.91	0.91	0.91	0.91	0.91	0.97	0.97	0.97	0.97	0.97
77°F (25.0°C)	TC	15896	15918	15935	15935	15957	16854	16874	16891	16894	16914	17861	17884	17904	17908	17928	18937	18958	18979	18982	19005
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	0.87	0.87	0.87	0.87	0.87	0.92	0.92	0.92	0.92	0.92	0.98	0.98	0.98	0.98	0.98	1.03	1.04	1.04	1.04	1.04
82°F (27.8°C)	TC	15900	15922	15939	15939	15961	16858	16878	16895	16898	16918	17866	17889	17909	17912	17932	18942	18963	18983	18986	19010
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	0.96	0.96	0.96	0.96	0.96	1.02	1.02	1.02	1.02	1.02	1.08	1.08	1.08	1.08	1.08	1.14	1.14	1.15	1.15	1.15
86°F (30.0°C)	TC	15898	15919	15936	15937	15959	16856	16876	16892	16896	16916	17863	17886	17906	17909	17929	18939	18960	18980	18983	19007
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	1.11	1.11	1.11	1.11	1.11	1.17	1.17	1.18	1.18	1.18	1.24	1.25	1.25	1.25	1.25	1.32	1.32	1.32	1.32	1.32
95°F (35.0°C)	TC	15892	15914	15931	15932	15955	16850	16870	16887	16890	16910	17857	17880	17900	17903	17923	18933	18953	18974	18977	19001
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81	0.92	0.57	0.67	0.76	0.76						

Climate 5000 Series - Gen 4

18K Btu Single Zone Regular System - 4-Way Cassette IDU



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Heating Performance - Indoor Dry Bulb Temperature at 447 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	8512	8342	8300	8267	8192	1.47	1.48	1.48	1.49	1.49
-4°F(-20.0°C)	10871	10653	10600	10558	10463	1.55	1.57	1.57	1.57	1.58
0°F(-17.8°C)	11873	11636	11578	11531	11428	1.61	1.63	1.63	1.63	1.64
5°F(-15.0°C)	13127	12864	12800	12749	12634	1.69	1.70	1.71	1.71	1.72
14°F(-10.0°C)	14665	14372	14300	14243	14115	1.79	1.80	1.80	1.81	1.82
17°F(-8.3°C)	15178	14874	14800	14741	14608	1.82	1.83	1.84	1.84	1.85
22°F(-5.6°C)	16047	15726	15648	15585	15445	1.85	1.86	1.87	1.87	1.88
27°F(-2.8°C)	16916	16578	16495	16429	16281	1.87	1.89	1.89	1.90	1.90
32°F(0.0°C)	17786	17430	17343	17274	17118	1.90	1.92	1.92	1.92	1.93
35°F(1.7°C)	18307	17941	17851	17780	17620	1.92	1.93	1.94	1.94	1.95
37°F(2.8°C)	18948	18569	18476	18402	18237	1.92	1.93	1.94	1.94	1.95
42°F(5.6°C)	20550	20139	20038	19958	19778	1.91	1.93	1.93	1.94	1.94
44.6°F(7.0°C)	21191	20767	20663	20580	20395	1.91	1.93	1.93	1.94	1.94
47°F(8.3°C)	22152	21709	21600	21514	21320	1.91	1.93	1.93	1.93	1.94
52°F(11.1°C)	21038	20617	20514	20432	20248	1.57	1.58	1.58	1.59	1.59
57°F(13.9°C)	19924	19526	19428	19350	19176	1.22	1.23	1.24	1.24	1.24
59°F(15.0°C)	19479	19089	18994	18918	18747	1.09	1.10	1.10	1.10	1.10
62°F(16.7°C)	18811	18434	18342	18269	18104	0.88	0.89	0.89	0.89	0.90
64.4°F(18.0°C)	18276	17910	17821	17750	17590	0.72	0.72	0.72	0.73	0.73
69.8°F(21.0°C)	17073	16732	16648	16581	16432	0.35	0.35	0.35	0.35	0.35
75.2°F(24.0°C)	15870	15553	15475	15413	15275	0.27	0.28	0.29	0.30	0.31

Bolded values were tested. All other values are simulated and for reference only.